Campus Sensor Monitoring and Alerts

FUNCTIONALITY DOCUMENT | Backend Task 1

1. Purpose

Build a backend service that ingests campus sensor readings, evaluates organizer-defined alert rules, and lets organizers manage the alerts that are generated.

2. Functional Requirements

- Accept individual sensor readings and batches of readings.
- Be designed to handle bursts of 10,000 readings in one second.
- Allow organizers to create, edit, enable, and disable alert rules.
- Evaluate rules against a single reading, such as temperature exceeding a threshold.
- Evaluate time-window rules, such as average temperature over ten minutes exceeding a threshold.
- Create an alert whenever a reading or time window satisfies a rule.
- Allow organizers to view, acknowledge, and resolve alerts.
- Provide a sensor simulator for testing.

3. Core Data

A sensor reading contains the sensor identifier, metric type, measured value, and timestamp. The brief's example uses these fields:

Field	Meaning
sensorId	Identifier of the reporting sensor (example: library-floor-2).
type	Metric being reported (example: temperature).
value	Numeric sensor reading (example: 34.5).
timestamp	Time at which the reading was recorded (example: 2026-10-12T10:30:00Z).

4. Required Backend Behaviour

Reading ingestion

Receive both single readings and batches. The service design must account for the specified 10,000-readings-in-one-second burst.

Alert-rule management

Organizers can create, edit, enable, and disable rules. Rules may evaluate an individual reading or a time-window aggregate.

Alert lifecycle

When a rule is satisfied, create an alert. Organizers must be able to view the alert, acknowledge it, and resolve it.

5. Required Testing and Submission

- Include a sensor simulator to generate readings for testing.
- Provide complete source code and instructions for installation and running the service.
- Provide a load-test report describing performance during a burst of 10,000 readings in one second.
- Explain the architecture and important throughput or reliability trade-offs.

6. Scope Boundary

This document lists only the functionality specified in the GDG TECH Wildcard Entry brief for Backend Task 1. A separate frontend, dashboard, authentication system, or other unlisted feature is not included as a requirement.